

HADOOP AND SPARK ARCHITECTURES

Hadoop and Spark, developed by the Apache Software Foundation, are widely used open-source frameworks for big data architectures.

Each framework contains an extensive ecosystem of open-source technologies that prepare, process, manage and analyze big data sets.

<https://www.ibm.com/think/insights/hadoop-vs-spark>

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APACHE HADOOP

- Apache Hadoop is an open-source software utility that allows users to manage Big Data sets by enabling a network of computers to solve large and intricate data problems.
 - It is a highly scalable, cost-effective solution that stores and processes **structured, semi-structured** and **unstructured data** (e.g., Internet clickstream records, web server logs, IoT sensor data, and others).
 - It is highly fault-tolerant and does not depend upon hardware to achieve high availability.
 - It is designed with a vision to look for the failures at the application layer.
 - It is a general-purpose form of distributed processing
 - It is built in Java, and accessible through many programming languages, for writing MapReduce code, including Python, through a Thrift client. It's available either open-source through the Apache distribution, or through vendors such as Cloudera, MapR, or HortonWorks.

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HADOOP ECOSYSTEM

- **Apache Hadoop** supports advanced analytics for stored data (e.g., predictive analysis, data mining, machine learning).
- It enables big data analytics processing tasks to be split into smaller tasks.
- The small tasks are performed in parallel by using an algorithm (e.g., MapReduce), and are then distributed across a Hadoop cluster (i.e., nodes that perform parallel computations on big data sets).
- The Hadoop ecosystem consists of four modules:
 - **Hadoop Distributed File System (HDFS)**: Primary data storage system that manages large data sets running on commodity hardware. It also provides high-throughput data access and high fault tolerance.
 - **Yet Another Resource Negotiator (YARN)**: Cluster resource manager that schedules tasks and allocates resources (e.g., CPU and memory) to applications.
 - **Hadoop MapReduce**: Splits big data processing tasks into smaller ones, distributes the small tasks across different nodes, then runs each task.
 - **Hadoop Common (Hadoop Core)**: Set of common libraries and utilities that the other three modules depend on.

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HADOOP

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| <ul style="list-style-type: none"> ■ Advantages: ■ Cost effective. ■ Processing operation is done at a faster speed. ■ Best to be applied when a company is having a data diversity to be processed. ■ Creates multiple copies. ■ Saves time and can derive data from any form of data. | <ul style="list-style-type: none"> ■ Disadvantages: ■ Can't perform in small data environments ■ Built entirely on java ■ Lack of preventive measures ■ Potential stability issues ■ Not fit for small data |
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APACHE SPARK

- Apache Spark, is an open source framework. It is a data processing engine for big data sets. Like Hadoop, Spark splits up large tasks across different nodes. However, it tends to perform faster than Hadoop and it uses random access memory to cache and process data instead of a file system. This enables Spark to handle use cases that Hadoop cannot.
 - initially developed in 2012, at the AMPLab at UC Berkeley.
 - It is focused on processing data in parallel across a cluster, but the biggest difference is that it works in memory.
 - It is designed to use RAM for caching and processing the data.
- Benefits of the Spark framework include the following:
 - A unified engine that supports SQL queries, streaming data, machine learning (ML) and graph processing
 - Can be 100x faster than Hadoop for smaller workloads (link resides outside ibm.com) via in-memory processing, disk data storage, etc.
 - APIs designed for ease of use when manipulating semi-structured data and transforming data

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SPARK ECOSYSTEM

- **Apache Spark**, the largest open-source project in data processing, is the only processing framework that combines data and artificial intelligence. This enables users to perform large-scale data transformations and analyses, and then run state-of-the-art machine learning and AI algorithms.
- The Spark ecosystem consists of five modules:
 - **Spark Core:** Underlying execution engine that schedules and dispatches tasks and coordinates input and output (I/O) operations.
 - **Spark SQL:** Gathers information about structured data to enable users to optimize structured data processing.
 - **Spark Streaming and Structured Streaming:** Both add stream processing capabilities. Spark Streaming takes data from different streaming sources and divides it into micro-batches for a continuous stream. Structured Streaming, built on Spark SQL, reduces latency and simplifies programming.
 - **Machine Learning Library (MLlib):** A set of machine learning algorithms for scalability plus tools for feature selection and building ML pipelines. The primary API for MLlib is DataFrames, which provides uniformity across different programming languages like Java, Scala and Python.
 - **GraphX:** User-friendly computation engine that enables interactive building, modification and analysis of scalable, graph-structured data.

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SPARK

■ Advantages :

- Perfect for interactive processing, iterative processing and event steam processing
- Flexible and powerful
- Supports for sophisticated analytics
- Executes batch processing jobs faster than MapReduce
- Run on Hadoop alongside other tools in the Hadoop ecosystem

■ Disadvantages :

- Consumes a lot of memory
- Issues with small file
- Less number of algorithms
- Higher latency compared to Apache fling

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HADOOP vs SPARK

- Hadoop reads and writes files to HDFS, Spark processes data in RAM using a concept known as a Resilient Distributed Dataset.
- Spark can run either in stand-alone mode, with a Hadoop cluster serving as the data source, or in conjunction with Mesos. In the latter scenario, the Mesos master replaces the Spark master or YARN for scheduling purposes.
- Spark is structured around Spark Core, the engine that drives the scheduling, optimizations, and RDD abstraction, as well as connects Spark to the correct filesystem (HDFS, S3, RDBMS, or Elasticsearch). There are several libraries that operate on top of Spark Core, including Spark SQL, which allows you to run SQL-like commands on distributed data sets, MLLib for machine learning, GraphX for graph problems, and streaming which allows for the input of continually streaming log data.

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HADOOP vs SPARK

- Performance
- Use Cases
- Latency
- Data
- Cost
- Algorithm Used
- Fault Tolerance
- Security
- Machine Learning
- Scalability
- Language support
- User-friendliness
- Resource Management